# Who are the Stanislaus monkeyflowers? Investigating *Erythranthe marmorata* and putative relatives in the central Sierra Nevada foothills

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The yellow monkeyflowers of the Western Sierra Nevada Foothills represent a mix of well-studied and understudied species. My work is delving into the morphology and genetics of the Stanislaus monkeyflower, *Erythranthe marmorata* (reclassified by Guy L. Nesom in 2012 by combining *Mimulus whipplei* and *Mimulus marmoratus*) and some of its putative close relatives from the Sierra Nevada foothills, listed below.

*Erythranthe marmorata,* Stanislaus monkeyflower (Greene 1895) Nesom 2012




Img. A: *E. marmorata* flowering calyx



Img. B: *E. marmorata* fertilized calyx



Img. C: *E. marmorata* growing in rock crevice, Knights Ferry, California

*Erythranthe guttata,* Seep monkeyflower (Fisher 1813) Nesom 2012



Img. D: *E. guttata* flowering calyx



Img. E: *E. guttata* flowering and fertilized calyxes



Img. F: *E. guttata* nonflowering voucher specimens

*Erythranthe pardalis,* Pennell's panther (Pennell 1974) Nesom 2012



Img. G (left): *E. pardalis* calyx detail

Img. E (right): *E. pardalis* growing beneath a rock face in Tuolumne County, California

photos by Mike Russler, 2025 for CalFlora



other sympatric monkeyflowers

*Erythranthe nasuta*
Shy monkeyflower
(Greene 1895) Nesom 2012

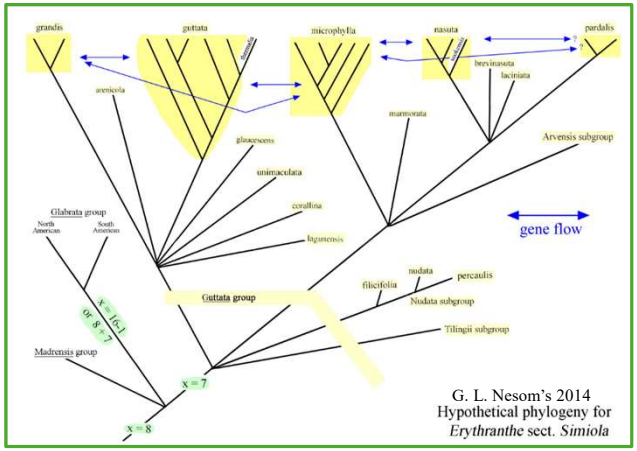
*Erythranthe laciniata*
Cutleaf monkeyflower
(Gray 1876) Nesom 2012

*Erythranthe arenaria*
Sand-loving monkeyflower
(Grant 1925) Nesom 2012

*Erythranthe tilingii*
Tiling's monkeyflower
(Regel 1869) Nesom 2012

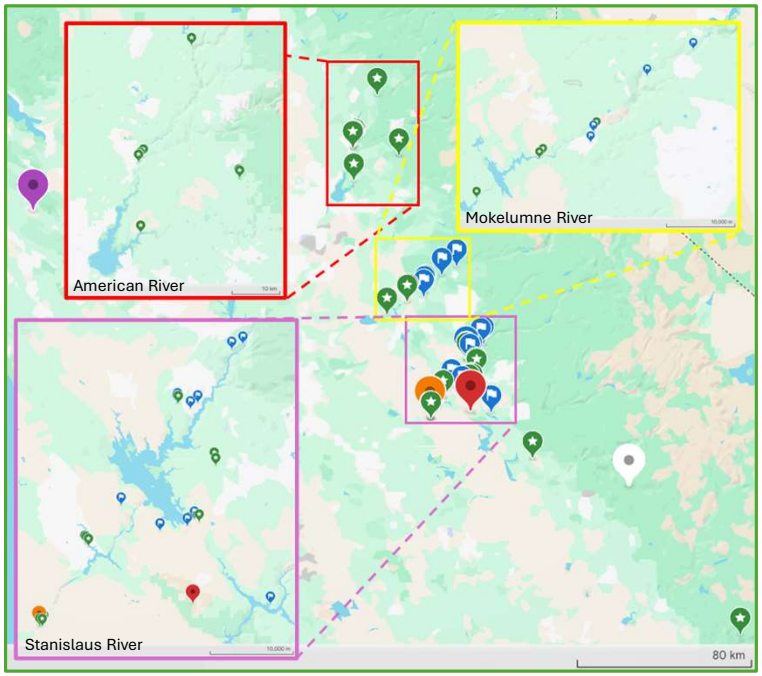
*Erythranthe microphylla*
Small-leaf monkeyflower
(Bentham 1846) Nesom 2012

*Erythranthe geniculata*
Bent-stem monkeyflower
(Greene 1885) Nesom 2011




G. L. Nesom's 2014 Hypothetical phylogeny for *Erythranthe* sect. *Simiola*

map of the central Sierra Nevada foothill region




Map Data: Google, 2026; Point Data: Nesom 2012, GBIF 2025, BLM 2026

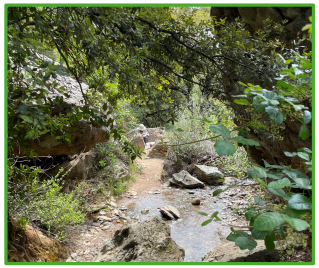
*E. marmorata* Observations:

Bureau of Land Management dataset

Global Biodiversity Information Facility

*Erythranthe sp.* Type Locations Within Area:

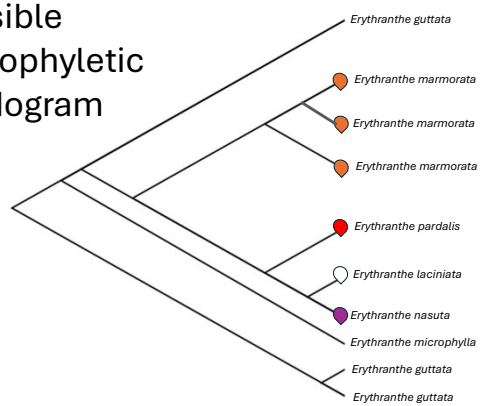
*E. marmorata*
*E. nasuta*
*E. pardalis*
*E. laciniata*



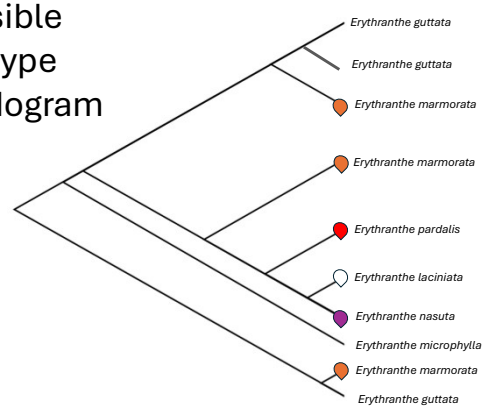
| | Spring 25 | Summer 25 | Fall 25 | Winter 25 | Spring 26 | Summer 26 | Fall 26 | Winter 26 | Spring 27 |
|---|---|---|---|---|---|---|---|---|---|
| Scouting | | | | | | | | | |
| Background Research | | | | | | | | | |
| Field Collection | | | | | | | | | |
| DNA Isolation | | | | | | | | | |
| Herbarium Survey | | | | | | | | | |
| Sequencing | | | | | | | | | |
| Analysis | | | | | | | | | |
| Composition | | | | | | | | | |

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Nesom 2012. Taxonomy of Erythranthe sect. Mimulosma (Phrymaceae). Phytoneuron 2012-41: 1–36. ISSN 2153 733X
Nesom 2014. Updated classification and hypothetical phylogeny of Erythranthe sect. Simiola (Phrymaceae). Phytoneuron 2014-81: 1–6. ISSN 2153 733X
GBIF.org (05 October 2025) GBIF Occurrence Download https://doi.org/10.15468/dl.882dqv
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CalFlora Photo Contributor: Mike Russler

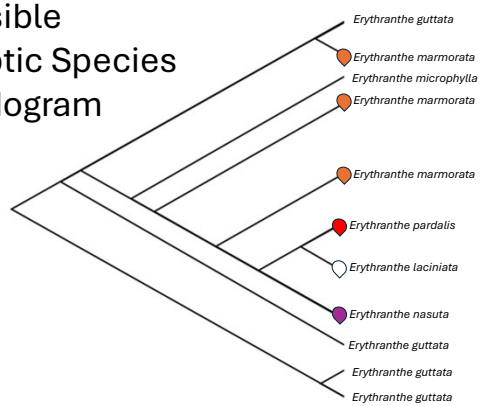
Possible Monophyletic Cladogram




Possible Ecotype Cladogram




Possible Cryptic Species Cladogram




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